

Advanced PC Disassembly and Component Analysis

1. Remove the side panel of a desktop PC case and take a clear photo showing the internal layout before any components are removed.

-> For your assignment, take **one clear photo of the inside of a desktop PC before removing any components**.

What the photo should show

- Motherboard
- CPU cooler
- RAM
- GPU, if installed
- HDD/SSD
- Power supply (PSU)
- Power/data cables
- Case fans
- PCIe slots and other visible motherboard components

Suggested caption

Photo – Internal PC Layout Before Disassembly: The side panel has been removed, showing the complete internal layout of the desktop PC before any components are disconnected or removed. All major components and their connections are visible.

2. Carefully remove the RAM module(s) from the motherboard, document the removal steps, and record the RAM type, speed, and form factor as printed on the module.

-> RAM Removal and Documentation

1. **Power off the PC** and unplug the power cable.
2. Remove the side panel and locate the **RAM modules in the DIMM slots** beside the CPU.

3. Touch the metal case or use an **anti-static wrist strap** before handling the RAM.
4. Push the **retaining clips** at the ends of the RAM slot outward.
5. Hold the RAM module **only by its edges** and gently lift it straight out.
6. Place the module in an **anti-static bag**.
7. Take a clear photo of the RAM label so its specifications can be read.

Record the specifications

Fill in the details printed on your module:

Specification	Your RAM
Type	DDR3 / DDR4 / DDR5
Speed	e.g., 3200 MHz
Form Factor	DIMM (desktop RAM)
Capacity	e.g., 8 GB
Module Label/Model	As printed on the RAM

Suggested photo caption:

RAM Module Removed: The RAM was carefully removed from the motherboard's DIMM slot by releasing the retaining clips and handling the module by its edges. The module label shows its RAM type, speed, capacity, and desktop DIMM form factor.

3. Disconnect and remove the storage drive (HDD or SSD) from the system, then list all connector types (SATA, power, etc.) used and describe how the drive was mounted inside the case.

Hint: Pay attention to both cable connectors and any brackets or screws securing the drive.

-> Storage Drive Removal and Documentation

1. **Shut down the PC** and disconnect the power cable.
2. Remove the case side panel and locate the **HDD/SSD**.
3. Identify and disconnect all cables connected to the drive.

4. For a **2.5-inch/3.5-inch SATA drive**:
 - **SATA data connector** → connects the drive to the motherboard.
 - **SATA power connector** → connects the drive to the PSU.
5. Remove the screws or release the mounting mechanism holding the drive.
6. Carefully slide/lift the drive out of its **drive bay, tray, or bracket**.
7. Place the drive in an **anti-static bag** and photograph it.

Connector and Mounting Record

Item	Typical connection/mounting
Data connector	SATA data cable
Power connector	SATA power cable from PSU
Mounting	Drive bay/tray or mounting bracket
Fasteners	Usually screws or a tool-less locking mechanism

Suggested assignment description:

Storage Drive Removal: The storage drive was disconnected from the motherboard and power supply before removal. The SATA data cable connected the drive to the motherboard, while the SATA power connector supplied power from the PSU. The drive was secured inside the case using a drive bay/tray and mounting screws. After removing the screws, the drive was carefully taken out and placed in an anti-static bag.

4. Identify and remove one expansion card (such as a graphics card or Wi-Fi card), then document the connector type (e.g., PCIe), screw type used, and any locking mechanisms you encountered.

-> Expansion Card Removal — Graphics Card Example

1. **Power off and unplug** the PC before opening the case.
2. Locate the **graphics card (GPU)** installed in the motherboard's PCIe slot.
3. If present, disconnect the GPU's **PCIe power cable** from the card.

4. Remove the **case/bracket screw** securing the GPU to the rear of the case. This is commonly a Phillips-head screw.
5. Locate the **PCIe slot retention clip** at the end of the motherboard slot and gently release it.
6. Hold the GPU by its edges and carefully pull it straight out of the PCIe slot.
7. Place the card in an **anti-static bag** and take a clear photo.

Document the Connections

Item	Observation
Expansion card	Graphics card (GPU)
Connector type	PCIe x16
Power connector	PCIe power connector, if required
Screw type	Typically Phillips-head case/bracket screw
Locking mechanism	PCIe slot retention clip
Mounting location	Rear expansion-slot bracket of the PC case

Suggested Assignment Description

Expansion Card Removal: I removed the graphics card from the motherboard's PCIe x16 slot. First, I disconnected its PCIe power cable, if present, and removed the Phillips-head screw securing its rear bracket to the case. I then released the PCIe slot retention clip and carefully removed the card by holding its edges. The graphics card was placed in an anti-static bag after removal.

5. Explain three safety precautions you followed during the disassembly process to prevent static electricity damage or injury, and describe how you ensured proper grounding.

Hint: Mention tools or accessories like anti-static wrist straps, and steps like unplugging the power supply.

-> **Safety Precautions During PC Disassembly**

1. **Unplugged the power supply:** I completely shut down the PC and disconnected the power cable before opening the case. This reduced the risk of electric shock and accidental component damage.
2. **Used an anti-static wrist strap:** I wore an **ESD/anti-static wrist strap** connected to an appropriate grounded metal point. This helped safely discharge static electricity from my body before handling sensitive components.
3. **Used a safe working surface:** I worked on a **clean, dry, non-carpeted surface** and placed removed components on an anti-static mat or in anti-static bags. I handled components by their edges and avoided touching electrical contacts.

Grounding Method

Grounding: I ensured proper grounding by using an anti-static wrist strap connected to a suitable grounded point while handling the components. I also kept the work area free from carpet and unnecessary static-generating materials. Before touching components, I made sure the PC was powered off and unplugged.